

The VPN Provisioning Portal

12

An executive overview of the workflows — every screen an operator touches, in order

workflows, one screen

Managed IPsec VPN service · Catalyst 8000v VTI + eBGP · Nautobot source of truth · Network-as-Code (Terraform) · Robot Framework validation

What the portal runs

The live service, as the portal itself reports it — every number below is read from the routers and from Nautobot, never typed in.

20

routers onboarded

3 headends · 4 branches · firewalls,
hosts and the interconnect

10 / 10

tunnels up

one static IPsec VTI per branch per
headend, each with eBGP

74

tests per change

the Robot Framework suites run at
the end of a job

~15 min

to add a branch

VM, onboarding, model, Terraform
apply and tests

The portal is the only place the service is changed.

An operator states intent on a form; the portal writes it to the intent document, seeds Nautobot, renders the Network-as-Code data from Nautobot, plans and applies Terraform against the routers, then runs the test suites and keeps the evidence. Nobody logs into a router to make a change.

One screen: “what do you want to do?”

The Provision tab opens on the task, not on a form. Each card says what the job does, how long it takes and who may start it.

BUILD

Add a branch · Add a headend

CHANGE A BRANCH

Change IKE authentication · Rotate a pre-shared key · Renew a certificate · Re-home
· Remove

THE WHOLE SERVICE

Deploy the model · Dry run · Run the tests ·
Edit the service model

CONFIGURATION

Check for drift · See what is running

Each card carries its own duration and the role it needs — five minutes to rotate a key, about fifteen to build a branch, and an approver before anything is removed.

VPN Provisioning Portal
C8000v IPsec VTI + eBGP · Nautobot source of truth · Network-as-Code (Terraform) · Robot Framework validation

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[approver](#) [sign out](#) · [dark](#)

What do you want to do? every change is a job: pick one, review what it will do, then watch it run on its own page

BUILD

- Add a branch**
A new customer site: its VM is created and bootstrapped, onboarded into Nautobot, then the branch and its headends are configured in one Terraform run.
guided wizard · about 15 minutes
- Add a headend**
A new ACME headend linked to every branch you choose: its VM, a WAN link and a tunnel per branch, then both sides applied.
about 20 minutes

CHANGE A BRANCH

- Change IKE authentication**
Move a branch between its pre-shared key and a certificate from the lab CA — profiles, keyring, trustpoint and the SAs follow.
about 6 minutes
- Rotate a pre-shared key**
A new key for one branch, pushed to it and to every headend that keys for it, then its SAs are cleared and the tunnels verified.
about 5 minutes
- Renew a certificate**
A fresh CSR signed by the lab CA and imported; the router re-authenticates every tunnel with the new certificate.
about 4 minutes
- Re-home a branch**
Change which headends a branch connects to: new links and tunnels are allocated, dropped ones removed everywhere.
about 12 minutes
- Remove a branch**
Decommission a branch: powered off, cleaned out of Nautobot, the intent, lab.conf and Terraform state, and its VM deleted.
approver · about 10 minutes

THE WHOLE SERVICE

- Deploy the model**
Push what the model says to every router: seed Nautobot, render the NaC data from it, plan and apply.
Nautobot → NaC → Terraform → Golden Config
- Dry run**
The same pipeline, stopping after the Terraform plan — what would change, and nothing more.
nothing is pushed
- Run the tests**
The Robot Framework suites against the live lab, ending with an evidence report.
about 55 minutes
- Edit the service model**
Site, VPN service, crypto profile, internet breakout, capacity and the routers table — everything a deploy pushes.
the form below

CONFIGURATION

- Check for drift**
Golden Config backs up every router, renders the intended configuration from Nautobot and compares them — then remediate or re-apply what drifted.
- See what is running**
Every job this portal has run, newest first, with its steps, its log and its test report.

Provision — the task launcher; every card is one job

Every job looks the same

Pick a task, review what it will do, then watch it run on its own page — steps, live log, tests, ticket.

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Nautobot: [open](#) · 20 routers onboarded ·
[admin](#) [approver](#) [sign out](#) · [dark](#)

[Jobs](#)

Change IKE authentication · spoke3
2026-09-23_13:55-28-475b
 [results](#)
[ticket CHG0042002](#)

SUCCESS · the branch moves between pre-shared key and certificate: profiles, keyring, trustpoint, SAs
success
 finished 2:21:39 PM

9 / 9
 steps completed

1571s
 took

admin
 started by

74 / 74
 tests passed

Steps
what the pipeline does, in order — each step carries what it did in one line

- Validate the authentication change (spoke, headends, profiles)**
success · 0s
 spoke3: psk -> certificate on 3 tunnel(s) to central-headend, east-headend, west-headend; central-headend keeps the certificate profile; east-headend keeps the certificate profile; west-headend keeps the certificate profile
- Record the spoke's authentication in the intent**
success · 0s
 spoke3: ike_authentication = certificate in lab-intent.json (the lab default; the override is dropped)
- Nautobot source of truth (seed)**
success · 174s
 seed complete: 9 changes:
- Render NAC data from Nautobot**
success · 4s
 nac/data/devices.nac.yaml + device_groups.nac.yaml regenerated from Nautobot
- Certificates: enrol the routers with the lab CA (trustpoint, CA cert, router cert)**
success · 35s
 5 router(s) hold a certificate from the lab CA; enrolled now: spoke3
- Terraform plan**
success · 99s
 Plan: 4 to add, 12 to change, 4 to destroy.
- Terraform apply + save config**
success · 361s
 2 added, 3 changed, 1 destroyed; 0 added, 9 changed, 0 destroyed; 2 added, 0 changed, 3 destroyed.
- Certificates: pre-shared keys off the routers, every tunnel re-authenticated with RSA**
success · 29s
 central-headend: 4/4 2 PSK, 2 RSA; east-headend: 3/3 1 PSK, 2 RSA; west-headend: 3/3 1 PSK, 2 RSA; spoke1: 2/2 2 PSK; spoke2: 2/2 2 PSK; spoke3: 3/3 3 RSA; spoke4: 3/3 3 RSA
- Robot Framework tests**
success · 867s
 74/74 tests passed

What that gives you

One job at a time — two changes can never race each other on the same router.

Every step names what it did: how many objects Nautobot changed, what Terraform added and destroyed.

A failed job resumes from the failed step; the successful steps are not repeated.

The tests are part of the job, not an afterthought — the evidence is attached to it.

Add a branch — who it is

discovered by Nautobot onboarding: role and ports follow the physical wiring

+ Add spoke

1 · Identity & metadata

2 · Addressing (auto-suggested)

3 · Review & provision

Hostname

spoke5

Management IP suggested

10.2.0.18

Region (picks the nearest headends)

East

Site (branch location in Nautobot) suggested

branch-5

City (placed on the map; stored as the Nautobot location's coordinates) suggested

Philadelphia, PA

Coordinates (lat / lon — filled in for a catalogue city)

39.9526

-75.1652

Site code

EA-5

Contact

noc-east@lab.local

IKE authentication

certificate (lab CA)

Pre-shared key generated

BVosOPgBljX8VjmerlXdCZLr

Change ticket

CHG0042002

Comments

e.g. branch office 3, opened 2026-09

Customer — every branch is a customer of ACME (a Nautobot tenant; the address becomes the branch location's physical address). A proposed company is filled in; change it freely.

Company proposed

Ironwood Stores Co.

Industry

Retail

Service tier

Silver

Street address

7208 Maple St, Suite 630, Philadelphia, PA 43682

ACME account · contract start

ACME-405091

09/29/2023

Connect to headends (select at least 2 — every spoke gets redundant headends)

☒ central-headend (Central, nearest · 4/50 tunnels · fw-central 32/50 Mbps · 2 free · ports Gi6, Gi7, Gi8)

☒ east-headend (East, nearest · 3/50 tunnels · fw-east 24/40 Mbps · 2 free · ports Gi5, Gi6, Gi7, Gi8)

The branch name and its customer come from the service model; the portal checks the name is free in Nautobot and in the lab.

Choosing two headends is what makes the branch dual-homed — the wizard will allocate a tunnel to each.

Nothing is created yet. Cancel costs nothing.

Add a branch — where it lives

Add a spoke router guided provisioning: VM → bootstrap → Nautobot → NAC/Terraform on hub + spoke → tests

Add a spoke router guided provisioning: VM → bootstrap → Nautobot → NAC/Terraform on hub + spoke → tests

1 · Identity & metadata

2 · Addressing (auto-suggested)

3 · Review & provision

Router-id (Loopback0) <i>suggested</i>	Site LAN /24 (Loopback10) <i>suggested</i>	AS number <i>suggested</i>
10.255.1.10	192.168.17.0/24	65207

Headend	Edge port <i>suggested</i>	Spoke port	WAN p2p /30	Tunnel #	Tunnel /30
central-headend via firewall fw-central · free: eth6, eth7, eth8	6	2	100.65.8.0/30	8	172.17.8.0/30
east-headend via firewall fw-east · free: eth5, eth6, eth7, eth8	5	3	100.65.12.0/30	12	172.17.12.0/30

Hub takes the first address of each /30, the spoke the second. The same TunnelN number is used at both ends. central-headend: 5 of 50 tunnels after, 40 of 50 Mbps on fw-central (2 slots before) · east-headend: 4 of 50 tunnels after, 32 of 40 Mbps on fw-east (2 slots before)

Cancel
Back
Next

Each headend line shows the free ports on its firewall and how many tunnel slots remain — capacity is part of the form.

Change any value and the portal re-validates it against the live lab, not against a stale copy.

Step 2 of 3 — every field suggested from the next free value, and editable

The last step is the whole change in one place: what will be created, on which devices, under which ticket.

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Add a spoke router guided provisioning: VM → bootstrap → Nautobot → NAC/Terraform on hub + spoke → tests

1 · Identity & metadata
2 · Addressing (auto-suggested)
3 · Review & provision

Spoke spoke5 (new VM, 10.2.0.18)

Customer	Ironwood Stores Co. · Retail · Silver · ACME-405091 7208 Maple St, Suite 630, Philadelphia, PA 43682
Design pattern	ACME-DH Dual headend (resilient) (2 tunnels)
central-headend	GigabitEthernet2 100.65.8.2/30 · Tunnel8 172.17.8.2/30 → 100.64.2.2 · neighbor 172.17.8.1 remote-as 65204 · ip route 100.64.2.2/32 -> 100.65.8.1 (fw-central)
east-headend	GigabitEthernet3 100.65.12.2/30 · Tunnel12 172.17.12.2/30 → 100.64.1.2 · neighbor 172.17.12.1 remote-as 65200 · ip route 100.64.1.2/32 -> 100.65.12.1 (fw-east)
Loopbacks	Loopback0 10.255.1.10/32, Loopback10 192.168.17.1/24
eBGP	AS 65207
Internet	breakout enabled – every firewall's eth9 on the NAT network; headends originate a default, branches prefer the nearest
Crypto	profile VPN-IPSEC (AES-256-CBC/SHA256/DH14); IKE authentication default certificates from the lab CA

Hub central-headend + firewall fw-central (coordinated configuration)

Firewall	fw-central eth6 100.65.8.1/30 (IKE / ESP / ICMP filter)
Tunnel	Tunnel8 172.17.8.1/30 · source 100.64.2.2 (shared WAN behind fw-central) → dest 100.65.8.2, ipsec profile
Route	ip route 100.65.8.0 255.255.255.252 -> 100.64.2.1 (fw-central)
eBGP	neighbor 172.17.8.2 remote-as 65207
Nautobot	cable fw-central eth6 ↔ GigabitEthernet2, VPN tunnel central-headend-spoke5, endpoints, peering
Capacity	5 / 50 tunnels after provisioning · 40 / 50 Mbps on fw-central (1 aggregate slots left)

One button, one job

The change ticket is required: it travels with the job and appears in the audit trail.

Provisioning is one job — VM, onboarding, Nautobot, Terraform on both ends, then the tests.

About fifteen minutes later the branch is carrying traffic and the tests say so.

4

Add a branch — watch it run

The job page streams each step as it completes, with what that step actually changed.

Deployment status

2026-09-24_12-02-23-185d · [open the job page](#)**SUCCESS** · plan · by admin · started 12:02:23 PM · finished 12:06:16 PM (232s)

- **Validate intent** success · 0s
20 devices, 10 tunnels, VPN IPSEC_VPN
- **Save intent** success · 0s
lab-intent.json written
- **Nautobot source of truth (seed)** success · 176s
seed complete: 0 changes (nothing new)
- **Render NAC data from Nautobot** success · 4s
nac/data/devices.nac.yaml + device_groups.nac.yaml regenerated from Nautobot
- **Terraform plan** success · 52s
No changes. Your infrastructure matches the configuration.

[Log](#)

Recent runs

Started	Mode	Status	VPN	Ticket	Tests
9/24/2026, 12:02:23 PM	plan by admin	success	IPSEC_VPN	CHG0042002	
9/24/2026, 9:40:44 AM	golden by scheduler	success	IPSEC_VPN	CHG0042002	
9/24/2026, 3:40:21 AM	golden by scheduler	success	IPSEC_VPN	CHG0042002	
9/23/2026, 9:41:00 PM	plan by operator	success	IPSEC_VPN	CHG0042002	
9/23/2026, 9:33:45 PM	auth spoke1 by operator	success	IPSEC_VPN	CHG0042002	
9/23/2026, 9:27:44 PM	auth spoke1 by operator	success	IPSEC_VPN	CHG0042002	
9/23/2026	plan by	failed	IPSEC_VPN	CHG0042002	

VPN service

Nautobot core VPN app: VPN → tunnels → hub/spoke endpoints

VPN name

Change ticket

Owner

Description

IPSEC_VPN

CHG0042002

network team

Hub-and-spoke IPsec VTIs with eBGP (one AS p

Tunnels (one static IPsec VTI per spoke)

Hub	Spoke	Hub port	Spoke port	WAN p2p /30	Tunnel #	Tunnel /30
-----	-------	----------	------------	-------------	----------	------------

The pipeline: validate intent → save intent → seed Nautobot → render NaC → Terraform plan → apply → tests

Recent runs sit beside it: mode, who started it, the ticket and the outcome.

Add a headend

The same shape of form for the other direction: a new ACME headend, linked to every branch you choose.

Add a headend

A new ACME headend: it is registered in lab.conf, its VM is created and bootstrapped, onboarded into Nautobot, linked to every branch you tick below (a WAN port and a tunnel each), and then both sides are configured in one Terraform run. Every value is suggested and editable.

Hostname	Region	City	Management IP (OOB)
hub1	East	Philadelphia, PA	10.2.0.18
AS number	Router-id (Loopback0)	Site LAN (/24)	Site name
65207	10.255.1.10	192.168.17.0/24	hub1-hq
Site code	Contact	Change ticket	RAM (MiB)
EA-HQ	noc-east@lab.local	CHG0042002	4096

Branches to connect each one gets a WAN link to this headend and a tunnel; they are re-defined for the new port and rebooted

- ☒ spoke1 ☒ spoke2 ☒ spoke3 ☒ spoke4
- ☒ Golden Config backup + compliance afterwards
- ☐ run the Robot Framework suites afterwards (≈15 min)

The other direction

A headend is the heavier job (about twenty minutes): a new router, a new firewall link, and a tunnel added at both ends of every branch it serves.

The wizard allocates the tunnel numbers and /30s for each branch at once, so the mesh stays consistent.

The same pipeline runs it, and the same tests prove it.

east-hq	65200	10.255.1.	192.168.1	PSK + certificate	361 d	until 2027-09-21		Renew cert...
west-hq	65206	10.255.1.	192.168.1	PSK + certificate	361 d	until 2027-09-21		Renew cert...

Every value suggested — identity, WAN link, firewall ports and a tunnel per branch

9

Change a branch, safely

Day-two work is a short job with a narrow blast radius — one branch, one property, one ticket.

discovered by Nautobot onboarding; role and ports follow the physical wiring

IP

Role

0.15

hub

0.11

hub

0.17

hub

0.12

spoke

0.13

spoke

0.14

spoke

0.16

spoke

0.19

dci

0.20

partner

0.22

firewall

0.21

firewall

0.22

firewall

Change IKE authentication spoke1

Authenticate with

certificate from the lab CA

Spoke spoke1 (10.2.0.12)

Now pre-shared key

Wanted certificate — IKEv2 profile
VPN-IKEV2 (lab default)

On the spoke RSA key pair, trustpoint
with the pinned CA
fingerprint, CA
certificate, router
certificate enrolled;
keyring and pre-share
profile removed

Tunnels Tunnel1 → east-headend
Tunnel4 → central-headend

Headend central-headend

Profile already serves certificate
spokes

Tunnel re-pointed at the certificate
IPsec profile; re-authenticates

Headend east-headend

Profile already serves certificate
spokes

Tunnel re-pointed at the certificate
IPsec profile; re-authenticates

Order: intent → Nautobot (the tunnels move to the certificate VPN profile) → NaC render → certificates enrolled /
retired → staged terraform (new profiles first, tunnels re-pointed, old profiles last) → IKEv2 SAs re-authenticated and
verified as modelled → Golden Config. Each tunnel of the spoke is down for a few seconds.

nothing is pushed

about 55 minutes

the form below

See what is running

Every job this page
first, with its step
report.

Jobs page

Rotate a pre-shared key

Which branch gets a new key? Its headends are re-keyed in the same run.

spoke1

key rotated 2026-09-20

spoke3

key rotated 2026-09-22

Cancel

Site code

LAB-IPSEC

Contact

noc@lab.local

Description

C8000v IPsec V

Moving a branch to certificates re-enrols it with the CA, swaps the IKEv2 profile, keyring and trustpoint, clears the security associations and waits for every tunnel to re-authenticate — nine steps, and the tests at the end.

Rotating a pre-shared key is the same idea in five minutes: a new key on the branch and on every headend that serves it, then the SAs are cleared and the tunnels verified.

Remove a branch

Decommissioning says what it is about to delete before it deletes anything — and needs an approver.

Remove spoke spoke4

Spoke spoke4 (10.2.0.16)

VM	powered off, then undefined
Nautobot	device, interfaces, addresses, prefixes, VPN endpoints, BGP instance, AS 65205
Released	LAN 192.168.15.0/24, router-id 10.255.1.6, WAN 100.65.2.0/30, tunnel 172.17.2.0/30, WAN 100.65.7.0/30, tunnel 172.17.7.0/30, WAN 100.65.11.0/30, tunnel 172.17.11.0/30

Hub east-headend + fw-east (coordinated clean-up)

Tunnel	Tunnel2 removed (Terraform destroy), static route removed
Firewall port	fw-east eth3 address removed, port disabled and freed
eBGP	neighbour for AS 65205 removed
Nautobot	VPN tunnel east-headend-spoke4, hub endpoint, cable, peering
Capacity	2 / 50 tunnels after removal · 16 / 40 Mbps on fw-east (3 aggregate slots after)

Guarded by design

Only a user with the approver role may start it; an operator sees the card but cannot run it.

The reverse of provisioning, in the same order: tunnels dropped at both ends, objects removed from Nautobot, lab.conf and Terraform state cleaned, the VM deleted.

The job and its log stay in the ledger afterwards.

08	10.25	Hub central-headend + fw-central (coordinated clean-up) Tunnel Tunnel7 removed (Terraform destroy), static route removed	Hub west-headend + fw-west (coordinated clean-up) Tunnel Tunnel11 removed (Terraform destroy), static route removed	
09	10.25			
	n/a			

The confirmation lists the router, the tunnels, the Nautobot objects and the Terraform state that will go

The ledger: every job the portal has run

Jobs is the operational record — what ran, against what, by whom, for how long, and whether the tests passed.

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Jobs

30 of 30 jobs — every change this portal makes is a job, and jobs execute one at a time

any status

any job

router, ticket, use

Refresh

0

running or queued

25

succeeded

4

failed

30

jobs kept

74

tests run in them

Recent jobs

newest first, live — click one to open its page, with its steps, its log and its test report

Started	Job	Target	Status	Progress	Duration	Started by	Ticket	Tests
9/24/2026, 12:02:23 PM	Dry run (plan only)	the whole lab	success	5/5	232s	admin	CHG0042002	Open
9/24/2026, 9:40:44 AM	Golden Config	the whole lab	success	1/1	57s	scheduler	CHG0042002	Open
9/24/2026, 3:40:21 AM	Golden Config	the whole lab	success	1/1	49s	scheduler	CHG0042002	Open
9/23/2026, 9:41:00 PM	Dry run (plan only)	the whole lab	success	5/5	223s	operator	CHG0042002	Open
9/23/2026, 9:33:45 PM	Change IKE authentication	spoke1	success	8/8	329s	operator	CHG0042002	Open
9/23/2026, 9:27:44 PM	Change IKE authentication	spoke1	success		338s	operator	CHG0042002	Open

The record

Thirty jobs kept · twenty-five succeeded · seventy-four tests run in them — the portal's own counters.

Every row carries its change ticket, so the record matches the change-management system.

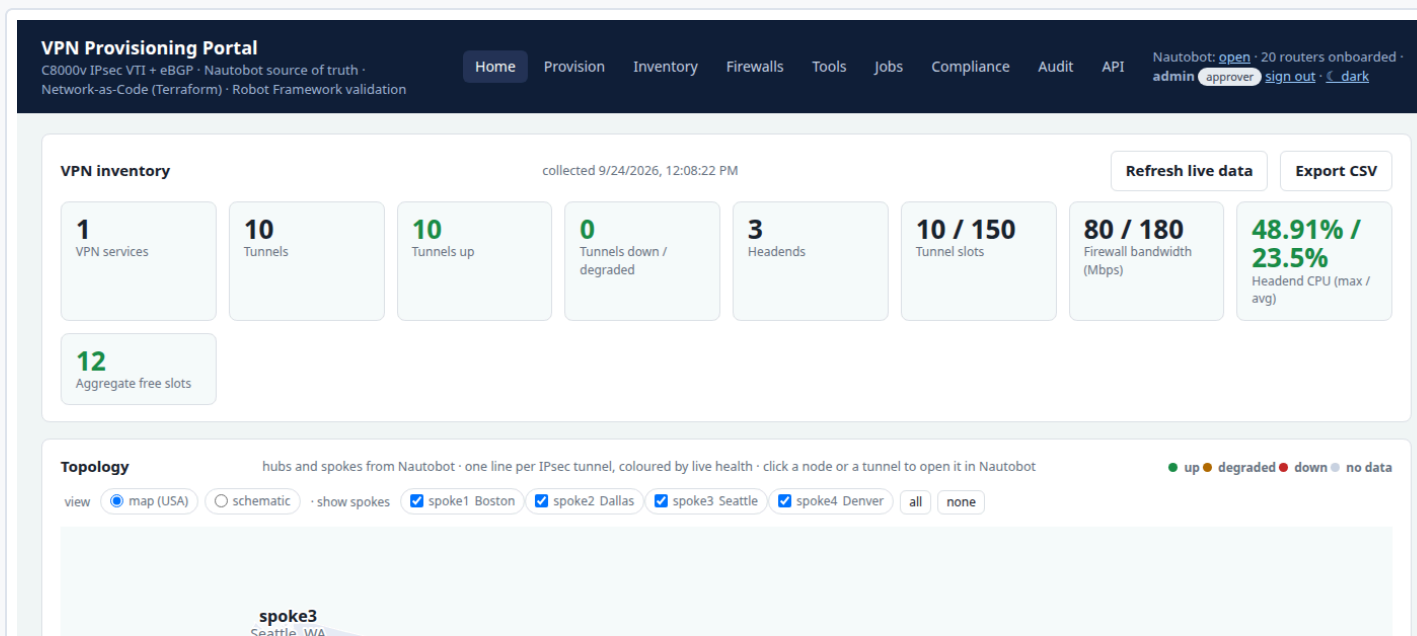
A failed job keeps its successful steps and offers Resume rather than a fresh start.

9/23/2026, 9:25:21 PM	Dry run (plan only)	the whole lab	failed	2/5	127s	operator	CHG0042002	Open	Resume
9/23/2026, 5:58:17 PM	Golden Config	the whole lab	success	1/1	57s	scheduler	CHG0042002	Open	
9/23/2026, 2:36:56 PM	Change IKE authentication	spoke3	success	9/9	1492s	admin	CHG0042002	Open	

Filterable by status, job type and target; a failed job offers Resume

The service as it stands

Home is the daily view: capacity and health first, the live topology below it.

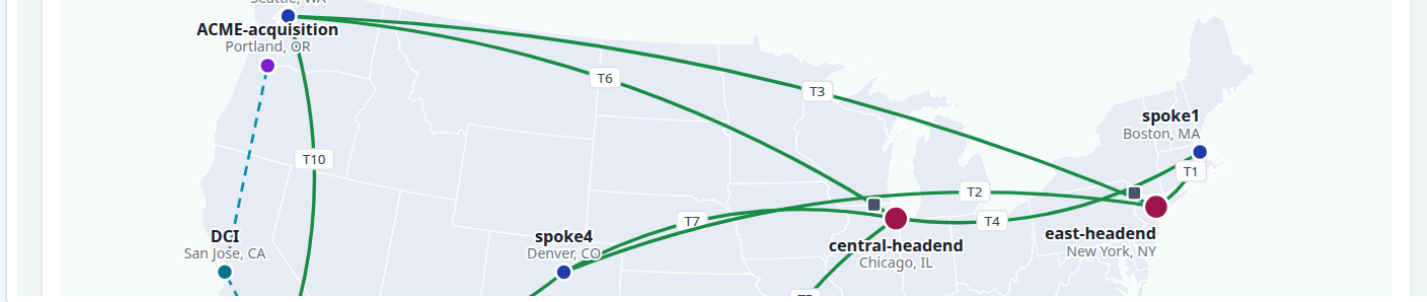


Capacity, not guesswork

Every tile is collected from the routers and the firewalls, with the time of collection shown; nothing is cached silently.

The topology is drawn from Nautobot and coloured live: a degraded tunnel is visible before anyone reports it.

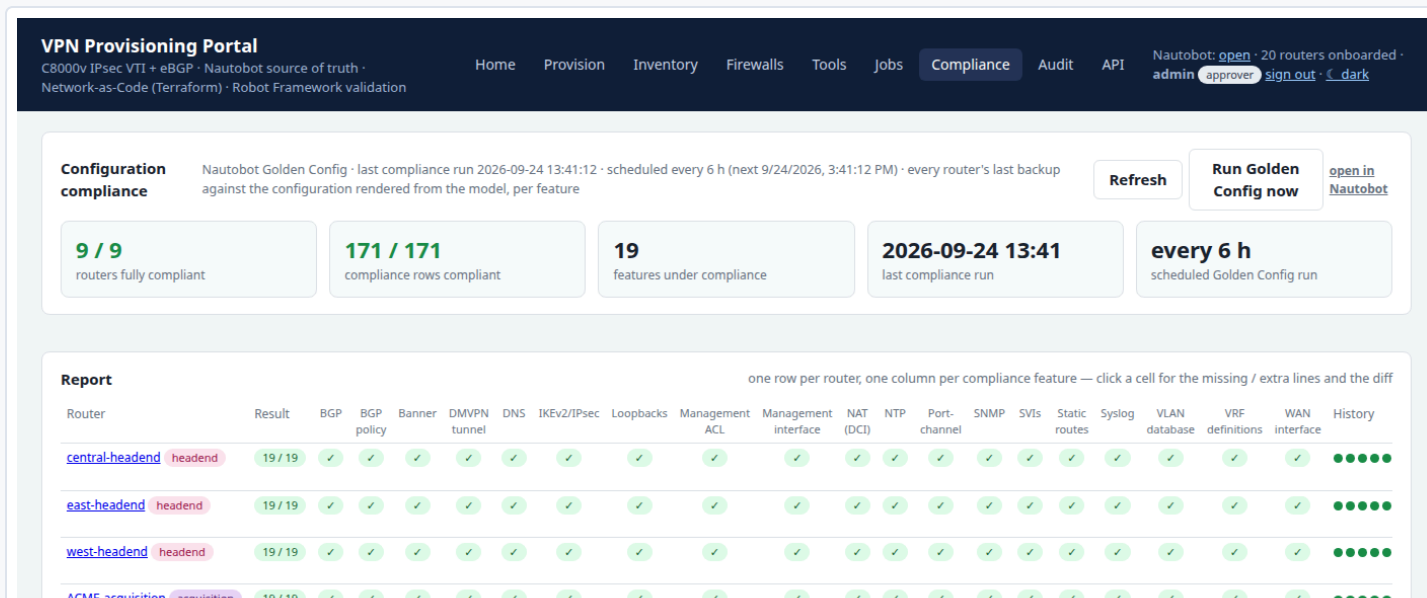
Free tunnel slots and spare firewall bandwidth are the numbers that decide whether the next branch fits.



Tunnels up, headend CPU, firewall bandwidth and free tunnel slots — then the map, coloured by live state

Drift, and what to do about it

Golden Config renders the intended configuration from Nautobot and compares it with what each router is actually running.



Two ways back

A scheduled run keeps the comparison current; the page shows when it last ran and what it found.

Remediate pushes only the difference; Re-apply pushes the whole intended configuration — both are ordinary jobs with logs and tests.

Drift history is kept, so 'when did this device start differing?' has an answer.

The portal is at <http://192.168.50.231:8090> — Provision to start a job, Jobs to see every one that has run, Home for the service as it stands. The full specification (115 requirements) is [docs/requirements.pdf](#).